Name: _____ ID: A

Part One- Multiple Choice **24 Points**

1. The function $f(x)=\left|x^2+\sin x\right|-\left|x^2-\sin x\right|$ is
   A. odd
   B. even
   C. neither odd nor even
   D. periodic

2. The domain of the function $f(x)=\ln\left(\frac{3}{x}-1\right)$ is:
   A. $(-\infty,3)$ B. $\mathbb{R}^+$ C. $(0,3)$ D. $\mathbb{R}$

3. The range of the function $f(x)=\frac{1}{|x|+3}$
   A. $[\frac{1}{3},\infty)$ B. $(-\infty,\frac{1}{3}]$ C. $(0,3]$ D. $(0,\frac{1}{3}]$

4. The smallest value of a such that the function $f(x)=2x^2+8x+3$ whrere $x\in[a,\infty)$ is one-to one is eqqual to
   A. -2 B. 2 C. 5 D. -5

when tanx is 1-1

5. $\tan\left[\tan^{-1}\left(\frac{52\pi}{7}\right)\right]=$
   A. $\frac{1}{7}\pi$ B. $\frac{2}{7}\pi$ C. $\frac{4}{7}\pi$ D. $\frac{3}{7}\pi$

6. $\lim\limits_{x\to 0^-}\frac{\sqrt{x^2}}{x}=$
   A. 0 B. -1 C. 1 D. does not exist

7. $\lim\limits_{x\to 0}\left(\frac{x}{\sin x+\tan 3x}\right)=$
   A. 1 B. $\frac{1}{3}$ C. $\frac{1}{2}$ D. $\frac{1}{4}$

8. If $\lim\limits_{x\to\infty}\left(\sqrt{x^2+ax}-x\right)=4$ then $a=$
   A. 4 B. 8 C. -4 D. -8

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1) $f(x) = |x^2 + \sin x| - |x^2 - \sin x|$

Remember

$\sin(-x) = -\sin x$

$f(-x) = |x^2 + \sin x| - |x^2 + \sin x|$

$f(-x) = -(|x^2 + \sin x| - |x^2 - \sin x|) = -f(x)$

Therefore, it's odd.

2) $f(x) = \ln\left(\frac{3}{x} - 1\right)$

$D_f = R(f^{-1}(x))$: إيجاد مجال f^{-1}

$f^{-1}(x) = \frac{3}{e^x + 1}$

$0 < e^x < \infty$

$1 < e^x + 1 < \infty$

$1 > \frac{1}{e^x + 1} > 0$

$3 > f^{-1}(x) > 0$

$D_f = R(f^{-1}) = (0, 3)$

$f(x) = \ln\left(\frac{3 - x}{x}\right)$

إيجاد D_f : توحيد مقامات

$= \ln(3 - x) - \ln(x)$

$h \pm g$

$D_h \cup D_g$

$3 - x > 0 \qquad x > 0$

$3 > x$

$D_f = (0, 3)$

3) $f(x) = \frac{1}{|x| + 3}$

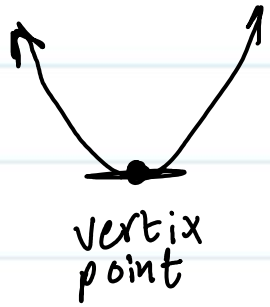
$0 \leq |x| < \infty$

$3 \leq |x| + 3 < \infty$

$1/3 \geq \frac{1}{|x| + 3} > 0$

$R(f) = (0, 1/3]$

4) $f(x) = 2x^2 + 8x + 3$



$$\text{vertix} = -\frac{b}{2a} = -\frac{8}{4} = -2$$

5) $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, $\tan\left(\tan^{-1}\left(5\frac{2\pi}{7}\right)\right)$

$$\begin{array}{r} 7 \\ 7\overline{)52} \\ -49 \\ \hline 3 \end{array}$$

$$\tan\left(\tan^{-1}\left(7\pi + \frac{3}{7}\pi\right)\right) = \frac{3}{7}\pi$$

6) $\lim_{x\to 0^-}\left(\frac{\sqrt{x^2}}{x}\right) = \lim_{x\to 0^-}\left(\frac{|x|}{x}\right) = \lim_{x\to 0^-}\left(-\frac{x}{x}\right) = -1$

7) $\lim_{x\to 0}\left(\frac{\sin x}{x} + \frac{\tan 3x}{x}\right)^{-1} = (1+3)^{-1} = 1/4$

8) $\lim_{x\to\infty}\left(\sqrt{x^2+ax} - x \cdot \frac{\sqrt{x^2+ax}+x}{\sqrt{x^2+ax}+x}\right)$

$= \lim_{x\to\infty}\left(\frac{x^2+ax-x^2}{\sqrt{x^2+ax}+x}\right) = \lim_{x\to\infty}\left(\frac{ax}{\sqrt{x^2+ax}+x}\right)$

$\lim_{x\to\infty}\left(\frac{ax}{|x|+x}\right) = \frac{a}{2} = 4$, $a = 8$

ame: _____

9. $\lim_{x\to\infty} e^{\left(\frac{x^3+x+1}{1+x^4}\right)} =$

A. 1 B. $-\infty$ C. ∞ D. 0

10. $\lim_{x\to 1^+} \left(-\ln(x-1)\right) =$

A. 1 B. 0 C. ∞ D. $-\infty$

11. The domain and range of the function $f(x) = \sin^{-1}(e^{-x})$ are (respectively)

A. $[0,\infty)$ and $(0,\frac{\pi}{2}]$

B. $(-\infty,0]$ and $[-\frac{\pi}{2},\frac{\pi}{2}]$

C. $[0,\infty)$ and $[-\frac{\pi}{2},\frac{\pi}{2}]$

D. $(-\infty,0]$ and $(0,\frac{\pi}{2}]$

12. If $f(x) = \frac{1}{1+4\tan^{-1}x}$ then $f^{-1}(x) =$

A. $\tan\left(\frac{1+x}{4x}\right)$

B. $\tan\left(\frac{1-x}{4x}\right)$

C. $\tan\left(\frac{1+x}{x}\right)$

D. $\tan\left(\frac{1-x}{x}\right)$

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9) $e^{\lim_{x\to\infty}\left(\frac{x^5+x+1}{1+x^4}\right)} = e^{\lim_{x\to\infty}\left(\frac{x^5}{x^4}\right)} = e^{\infty} = \infty$

10) $\lim_{x\to 1^+}\left(\ln\left(\frac{1}{x-1}\right)\right) = \ln\left(\lim_{x\to 1^+}\left(\frac{1}{x-1}\right)\right)$

$= \ln(\infty) = \infty$

11) $y = \sin^{-1}(e^{-x})$

$\sin^{-1}x$ $\qquad$ e^{-x}

R: $[-\pi/2, \pi/2] \cap (0, \infty) \longrightarrow (0, \pi/2]$

D: $[-1, 1] \cap \mathbb{R} \longrightarrow [0, \infty)$

D $(\sin^{-1}x)$ ↘

$-1 < e^{-x} < 1 \longrightarrow 0 < e^{-x} < 1$

ناسبة غير

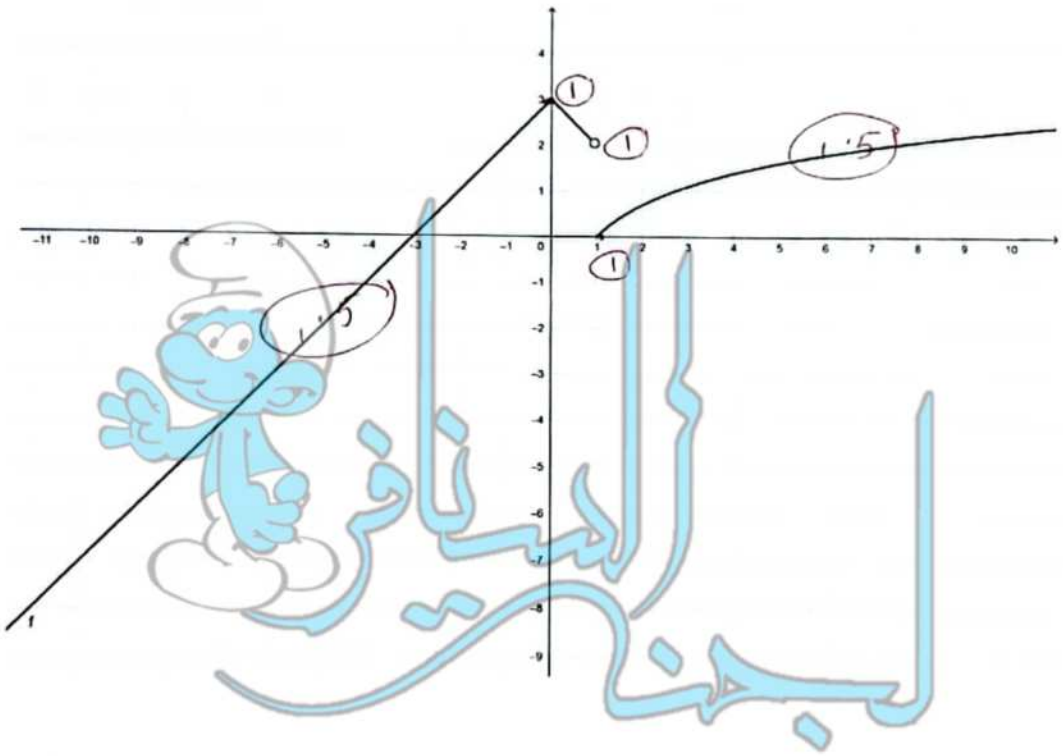
12) $x \not\equiv \frac{1}{1+4\tan^{-1}y}$ ، $4\tan^{-1}y + 1 = \frac{1}{x}$

$\tan^{-1}y = \frac{1}{4}\left(\frac{1}{x} - 1\right) = \frac{1-x}{4x}$ ، $y = \tan\left(\frac{1-x}{4x}\right)$

Part Two -Essay Question **(6 Points)**

Sketch The graph of the function f showing the main points and the $x-and\ y-intercepts$

$$f(x)=\begin{cases}3-|x| & , x<1\\ \ln x & , x\geq 1\end{cases}$$



Good Luck

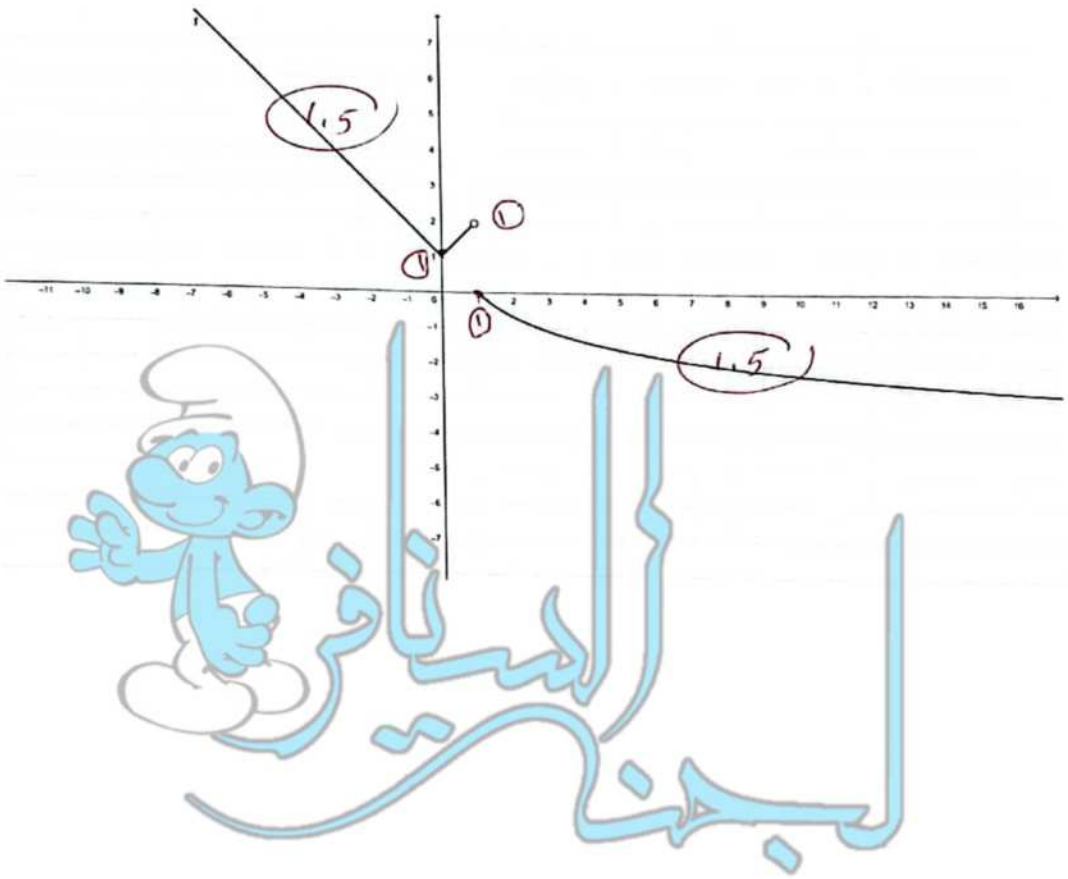
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Part Two -Essay Question **(6 Points)**

Sketch The graph of the function f showing the main points and the $x-and\ y-intercepts$

$$f(x)=\begin{cases} |x|+1 & , x<1 \\ -\ln x & , x\geq 1 \end{cases}$$



Good Luck

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